MAZEN BAHGAT

Applied AI Engineer

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PROFESSIONAL PROFILE

Applied AI Engineer and AWS Certified Cloud Practitioner with an MSc in Computing (AI-Native Software Systems) from Queen's University, building production Large Language Model (LLM) applications in Python. Experienced across the LLM integration layer with multiple model providers (OpenAI, Google Gemini, and self-hosted models via vLLM and Ollama), Retrieval-Augmented Generation (RAG) with LangChain, prompt engineering and evaluation, conversation memory, guardrails, and observability, tracing and evaluation with Langfuse, MLflow, Prometheus and Grafana. Delivers AI systems built for scalability, reliability, security and performance in production, using FastAPI, Docker, Kubernetes, CI/CD and AWS. Owns technical design decisions on team projects and has mentored 100+ engineering students.

PROFESSIONAL EXPERIENCE

DevOps/AI Intern | Rivora Labs, Egypt *Aug 2026 – Sep 2026*

- Owned the LLM integration layer for an agricultural monitoring platform, connecting the application to a self-hosted qwen3.5:9b model served through Ollama and the OpenWebUI API.
- Designed a secure FastAPI proxy so model URLs, credentials and system prompts are never exposed to the client, securing the platform's AI assistant.
- Implemented streaming chat, multimodal image analysis and per-conversation memory, with EXIF stripping on uploaded images.
- Built a full-stack dashboard in React, TypeScript, Python and FastAPI with Pydantic validation, Argon2 authentication, SQLite storage and Vitest tests.
- Built data pipelines for Sentinel-2 satellite imagery and weather APIs feeding a crop disease-risk model, and defined a PyTorch transfer-learning pipeline for disease classification.

AUTOSAR CEA Intern | Appraid Tech (in partnership with Vector), Egypt *Sep 2024 – Oct 2024*

- Completed AUTOSAR Certified Engineer (CEA) training on automotive software architecture using PREEvision, DaVinci Developer, DaVinci Configurator and CANoe.

Saher Radar Protection | Saudi Arabian technology company *Jun 2024 – Sep 2024*

- Built a multi-sensor radar-site security system (ATmega32, ESP32-CAM, SIM808) with event-triggered video capture and automated SMS and GPS alerts.

Professional Embedded Systems Intensive Training | NTI, Egypt *Mar 2024 – Jun 2024*

- Built an ADAS model car (adaptive cruise control, auto parking, lane tracking) with firmware-over-the-air (FOTA) updates over Wi-Fi on an STM32 microcontroller.

Robotics & Electronics Instructor | Armstrong, Egypt *Sep 2023 – Mar 2024*

- Mentored 100+ students across multiple cohorts and guided 30+ concurrent project teams from design to working hardware.

Embedded Controls Research Engineer | Arab Academy for Science and Technology, Egypt *Jun 2023 – Sep 2023*

- Designed a cascaded PI (position + velocity) controller for a linear servo mechanism with Wi-Fi telemetry, auto-generating production C code with MATLAB Embedded Coder.
- Published as: "Optimization of PI-Cascaded Controller's Parameters for Linear Servo Mechanism: A Comparative Study of Multiple Algorithms," IEEE Access, vol. 11, pp. 86377–86396, Aug 2023 (DOI: 10.1109/ACCESS.2023.3304333).

PROJECTS

HerHealthGPT: Multilingual Health Triage Assistant | MSc Capstone, MLOps/Systems Lead, Queen's University *2026*

- Co-built a live RAG triage assistant for women's health in English, French and Arabic (herhealthgpt.org); 25 of 30 testers preferred it.
- Bounded Qwen3.5-9B with closed vocabularies, Pydantic-validated JSON, six code gates and citation checks so it cannot diagnose (17 of 17 warning signs escalated); LangChain, FAISS and Qwen3-Embedding-4B retrieval lifted concern accuracy 5.19 points.
- Fine-tuned Qwen3.5-9B with QLoRA (Unsloth, PEFT), served it on vLLM and deployed on AWS with Terraform and CI/CD; our arXiv benchmark, HerHealthEval, showed fine-tuning cut clarification recall from 85.6% to 4.4%, so the base model stayed in production.

TestSeed: AI-Powered MongoDB Seed Data Generator | Queen's University *2026*

- Owned three features of an AI-native developer tool (AI seed generation, record-count controls, preview/inline editing), integrating the OpenAI API with constrained prompts, output validation and retry logic.
- Used Spec Kit to convert acceptance criteria into a structured specification, breaking a 5-user-story epic into 63 dependency-ordered tasks, then completed all 63 end-to-end with AI coding agents in 12 minutes 21 seconds.
- Debugged real AI generation failures (missing required fields, count mismatches, invalid ObjectId types) by feeding them back into stronger prompt constraints and a tightened validator.

Guarded RAG Tutor: Multi-Provider RAG with Guardrails | Queen's University *2026*

- Built a RAG pipeline in Python with LangChain text splitting, Sentence Transformers (all-MiniLM-L6-v2) embeddings and ChromaDB retrieval, turning user-uploaded PDFs into a searchable knowledge base.
- Integrated two model providers as guardrails: a Gemini 2.5 Flash scope classifier and the OpenAI Moderation API screen queries before and after generation, blocking 100% of off-topic and harmful test queries.
- Deployed an interactive Gradio interface for real-time document upload, querying and moderated responses.

LLM Evaluation, A/B Testing & Drift-Gated Rollout | Queen's University *Summer 2026*

- Ran MLflow-tracked few-shot prompt experiments (1/3/5-shot) on Qwen2.5-1.5B served by vLLM, reaching F1 = 1.000 at 3-shot; quantized the champion to 4-bit GPTQ for 3.6x lower latency and 65% smaller size, registered with champion/challenger aliases in the MLflow Model Registry.

- Built a FastAPI A/B traffic router (80/20 split, 250 predictions) across the fp16 and GPTQ models, with Evidently AI monitoring for prediction drift between them (drift score 0.0).
- Deployed to k3d via Argo Rollouts canary (20%→50%→80%→100%) gated by a Prometheus AnalysisRun querying sentiment_drift_score < 0.5, V2 auto-promoted after 3 consecutive passes; visualised on a Grafana drift-score dashboard with Pushgateway integration.

Cyber SOC Chatbot: Fine-Tuned LLM Deployed on AWS | Queen's University *May 2026*

- Provisioned end-to-end AWS infrastructure using Terraform IaC (VPC, EC2, S3, EMR, IAM) and preprocessed 114M+ cybersecurity records via PySpark on AWS EMR, balancing to 150K training samples.
- Fine-tuned TinyLlama-1.1B using QLoRA/PEFT (4-bit NF4) via Unsloth, reducing training loss by 58% in 41 minutes on a single GPU, and deployed it to AWS EC2 via the Ollama REST API.
- Containerized the full serving stack using Docker and OpenWebUI, configured with systemd auto-restart and least-privilege IAM roles following DevSecOps best practices.

CI/CD, GitOps & Observability Platform | Queen's University *Summer 2026*

- Built a 7-step CI pipeline on Gitea Actions for a FastAPI ML inference API: Ruff, Mypy, pytest (100% coverage), multi-stage Docker build, Git SHA image tagging and Syft SBOM generation.
- Deployed to Kubernetes (k3d) via GitOps with ArgoCD and Kustomize and Argo Rollouts canary releases, and built a Grafana dashboard on Prometheus metrics for request rate, 5xx rate, p50/p95/p99 latency and pod resources.

Portfolio Link: https://drive.google.com/file/d/1_McSX5Jai0DLpJoLd1LbKaMZ4IQe35xq/view?usp=sharing

EDUCATION

Master of Science in Computing (AI-Native Software Systems) | Queen's University, Canada *Sep 2025 – Sep 2026*

Delivered through the Digilians Initiative, including the 4-month NTI DevOps Track and 100+ hours of leadership and communication training.

Relevant Courses: Generative AI, AI & Machine Learning, Software Engineering with AI, DevOps & MLOps, Cloud Computing, Advanced Data Analytics

B.Eng. Mechatronics Engineering | Arab Academy for Science and Technology, Egypt *Oct 2018 – Jul 2023*

GPA: 3.5 / 4.0

TECHNICAL SKILLS

AI / LLM Engineering: LLM integration (OpenAI API, Google Gemini API, vLLM, Ollama, OpenWebUI API), LangChain, Retrieval-Augmented Generation (RAG), ChromaDB/FAISS, Sentence Transformers, prompt engineering, few-shot prompting, LLM evaluation, Langfuse, guardrails, content moderation, PII redaction (Presidio), streaming responses, conversation memory

Programming Languages: Python, TypeScript, JavaScript, SQL, Bash/Shell, C, C++, Embedded C, Java, YAML/JSON

AI / ML Deployment: PyTorch, HuggingFace Transformers, QLoRA/PEFT, Unsloth, GPTQ quantization, GGUF, ONNX Runtime, Ollama, vLLM, model containerization, REST API serving

MLOps: MLflow (experiment tracking, model registry), DVC, Evidently AI (drift monitoring), A/B traffic routing, champion/challenger promotion, Prometheus Pushgateway, AnalysisRun gating

Python & Web: FastAPI, Pydantic v2, Uvicorn, React 18, Next.js, Vite, TanStack Query, Vitest, Testing Library

Cloud / DevOps: Docker, Kubernetes, k3d, Helm, ArgoCD, Argo Rollouts, Kustomize, GitOps, CI/CD, GitHub Actions, Jenkins, Terraform, CloudFormation, Ansible, Infrastructure as Code, Prometheus, Grafana, Linux

AWS Services: EC2, S3, RDS, ECR, Lambda, API Gateway, DynamoDB, VPC, IAM, ECS, EKS, CloudWatch, CloudTrail, SNS, SES, Route 53, Secrets Manager, SSM, EBS, EMR

AI-Assisted Engineering: Spec Kit (spec-driven development), Superpowers, Cursor, Codex, ChatGPT, Claude Code, MCP (Model Context Protocol) integration: shadcn, Context7, MongoDB, next-devtools

Databases / Tools: PostgreSQL, MongoDB, SQLite, Git, GitHub, Jupyter, MATLAB, VS Code

Testing & Agile: pytest, Ruff, Mypy, SBOM generation (Syft), software testing, HIL/SIL, JIRA (Agile/Scrum)

CERTIFICATIONS

- AWS Certified Cloud Practitioner - Amazon Web Services - Oct 2026
- Git Basics for DevOps - KodeKloud - Dec 2025
- DevOps Prerequisite Course - KodeKloud - Dec 2025
- Docker for Beginners with Hands-on Labs - KodeKloud - Nov 2025
- Docker Basics for DevOps - KodeKloud - Nov 2025
- Fundamentals of Red Hat Enterprise Linux - Red Hat - Nov 2025
- Introduction to Git and GitHub - Google - Oct 2025
- AUTOSAR CEA - Vector & Appraid Tech - Sep 2024
- Introduction to AI - ITI - Jul 2022
- Fundamentals of Deep Learning - NVIDIA - Jul 2022
- Machine Learning Diploma - IMP - Jan 2021

VOLUNTARY & LEADERSHIP EXPERIENCE

Enactus Vice President | Arab Academy for Science and Technology *Jun 2021 – Oct 2022*

Student Union Vice President | Arab Academy for Science and Technology *Oct 2020 – Jun 2021*

LANGUAGES

Arabic (Native) - English (Fluent) - French (Beginner) - German (Beginner)